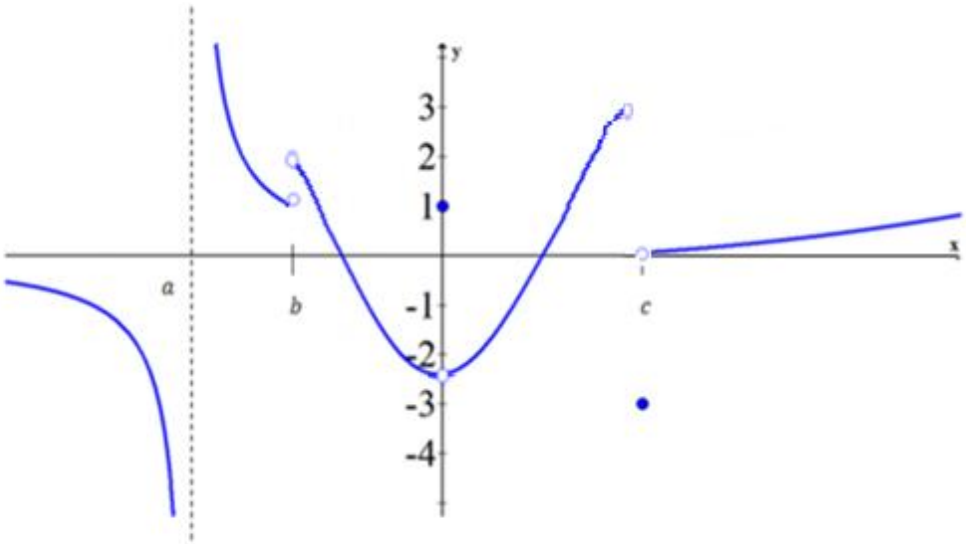
MTH 200 **Chapter 2: Limits**

2.2 Limits (An Intuitive Approach) Worksheet

1. Consider the following graph of the function $f(x)$. Determine each of the following limits.



a) $f(0)=1$

b) $\lim_{x\to 0^-} f(x)=-2.5$

c) $\lim_{x\to 0^+} f(x)=-2.5$

d) $\lim_{x\to 0} f(x)=-2.5$

e) $f(c)=-3$

f) $\lim_{x\to c^-} f(x)=3$

g) $\lim_{x\to c^+} f(x)=0$

h) $\lim_{x\to c} f(x)$ *DNE*

i) $f(a)$ *undefined*

j) $\lim_{x\to a^-} f(x)=-\infty$

k) $\lim_{x\to a^+} f(x)=\infty$

l) $\lim_{x\to a} f(x)$ *DNE*

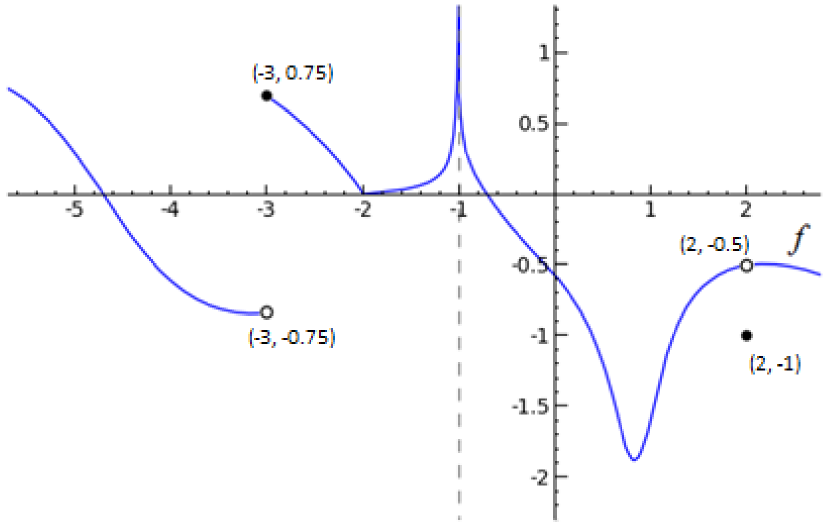
m) $f(b)$ *undefined*

n) $\lim_{x\to b^-} f(x)=1$

o) $\lim_{x\to b^+} f(x)=2$

p) $\lim_{x\to b} f(x)$ *DNE*

2. Consider the following graph of the function $g(x)$. Determine each of the following limits.



a) $\lim_{x\to-3} f(x)$ *DNE*

b) $f(2) = -1$

c) $\lim_{x\to-1^+} f(x) = \infty$

d) $f(-3) = 0.75$

e) $\lim_{x\to2} f(x) = -0.5$

f) $f(-1)$ *undefined*

g) $\lim_{x\to-1} f(x) = \infty$

h) $\lim_{x\to-3^-} f(x) = -0.75$

i) $\lim_{x\to-2} f(x) = 0$

3. Sketch the graph of an example of a function *f* that satisfies all of the given conditions:

$\lim_{x\to0^-} f(x) = 1,\quad \lim_{x\to0^+} f(x) = -1,\quad \lim_{x\to2^-} f(x) = -\infty,\quad \lim_{x\to2^+} f(x) = 1,\quad f(2) = 1,\quad f(0)$ *is undefined*